

Unitary Representations of the Galilei Group

In non-relativistic physics, inertial frames are related by Galilean transformations. If two observers are in inertial frames, their spacetime coordinates are related by

$$(\mathbf{x}, t) \mapsto (R\mathbf{x} + \mathbf{u}t + \mathbf{a}, t + s)$$

These transformations form a continuous group: the Galilei group. Since the laws of non-relativistic physics are invariant under these transformations, any quantum theory describing such physics must realize this symmetry. This is the core principle underlying our correspondence rules, whereby we identify particular dynamical variables with particular operators.

By Wigner's theorem, any linear symmetry transformation on the space of quantum states must be induced by a unitary operator acting on the underlying Hilbert space. Galilean transformations are continuous and connected to the identity, so they are represented by unitary operators. Thus for each Galilean transformation g there is a unitary operator $U(g)$ such that $U(0) = \mathbb{1}$. Every continuous one-parameter subgroup gives a unitary family

$$U(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon K\right) \quad (1)$$

where K is a self-adjoint operator called the generator of the transformation. For the Galilei group, the generators are:

Transformation	Unitary Operator	Generator
Spatial translation ($\mathbf{a}$)	$U(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\mathbf{a} \cdot \mathbf{p}\right)$	$\hat{p}_i$ (momentum)
Time translation (s)	$U(s) = \exp\left(\frac{i}{\hbar}sH\right)$	$-H$ ($-$ Hamiltonian*)
Rotation ($R(\boldsymbol{\theta})$)	$U(R) = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta} \cdot \mathbf{J}\right)$	J_i (angular momentum)
Galilean boost ($\mathbf{u}$)	$U(\mathbf{u}) = \exp\left(-\frac{i}{\hbar}\mathbf{u} \cdot \mathbf{G}\right)$	G_i (boost generator)

*According to Eq. (1), for the purpose of this discussion, the generator of time translation is $-H$ since H conventionally denotes the Hamiltonian as it appears in the Schrödinger equation. When a system is invariant under time translation, unitary time evolution gives $e^{isH}|\psi(t)\rangle = |\psi(t-s)\rangle$, which means the transformed state, at time t , is the same as the original state at time $t-s$. In other words, the transformed state indeed represents a time translation from t to $t+s$, in accordance with the group transformation listed above.

Our goal is to capture the group structure in the operator representations. Any sequence of space–time transformations is equivalent to another transformation in the group, so the composition of unitary transformations must respect this property. In fact, since physical states are only defined up to an overall phase, the product of two unitary operators implementing group elements may differ from the operator for the composed group element only by a phase. Generally speaking, this means

$$U(g_1)U(g_2) = e^{i\omega(g_1, g_2)}U(g_1g_2)$$

for some real-valued function $\omega(g_1, g_2)$. In many cases, we can determine the phase by additional considerations. Curiously, we cannot simply choose $\omega = 0$ in all cases.

Consider the following product of two infinitesimal unitary operators and their inverses:

$$W(\epsilon) = e^{-\frac{i}{\hbar}\epsilon K_\mu} e^{-\frac{i}{\hbar}\epsilon K_\nu} e^{\frac{i}{\hbar}\epsilon K_\mu} e^{\frac{i}{\hbar}\epsilon K_\nu} = \mathbb{1} - \frac{\epsilon^2}{\hbar^2}[K_\mu, K_\nu] + O(\epsilon^3) \quad (2)$$

which can be verified by expanding each operator to second order in ϵ . The operator product, $W(\epsilon)$, must be a symmetry transformation, so its generator must be a linear combination of the other generators of the group. The commutator can be written as

$$[K_\mu, K_\nu] = i\hbar \sum_\lambda c_{\mu\nu}^\lambda K_\lambda + ib_{\mu\nu}\mathbb{1}$$

where $c_{\mu\nu}^\lambda$ and $b_{\mu\nu}$ are real-valued coefficients, with the second factor contributing to the overall phase mentioned earlier:

$$W(\epsilon) = \exp\left(-\frac{\epsilon^2}{\hbar^2}[K_\mu, K_\nu]\right) + O(\epsilon^3) = \exp\left(-\frac{i\epsilon^2}{\hbar} \sum_\lambda c_{\mu\nu}^\lambda K_\lambda\right) \exp\left(-\frac{i\epsilon^2}{\hbar^2}b_{\mu\nu}\right) + O(\epsilon^3),$$

Observe that in order to capture the possibility of a global phase, the commutator must include a term proportional to the identity. Despite the generality of this discussion, we can apply these principles to each specific pair of generators.

Evaluation of commutators

Recall from our [theory of rotations](#), for any operator-valued 3-vector $\mathbf{A}$,

$$[J_i, A_j] = i\hbar\epsilon_{ijk}A_k \quad (3)$$

This ultimately has the effect of defining several phases discussed earlier. Additionally, we fix the following commutators:

$$[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0, \quad \text{and} \quad [\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij} \quad (4)$$

From the Jacobi identity for the triplet $(J_1, G_2, \hat{p}_1)$,

$$[J_1, [G_2, \hat{p}_1]] + [G_2, [\hat{p}_1, J_1]] + [\hat{p}_1, [J_1, G_2]] = 0,$$

$$[\hat{p}_1, G_3] = 0, \quad \text{and likewise, } [\hat{p}_i, G_j] = 0 \text{ for } i \neq j$$

From the Jacobi identity for $(J_1, G_2, \hat{p}_3)$,

$$[J_1, [G_2, \hat{p}_3]] + [G_2, [\hat{p}_3, J_1]] + [\hat{p}_3, [J_1, G_2]] = 0,$$

$$i\hbar[G_2, \hat{p}_2] + i\hbar[\hat{p}_3, G_3] = 0 \Rightarrow [G_2, \hat{p}_2] = [G_3, \hat{p}_3],$$

and by extension, $[G_1, \hat{p}_1] = [G_2, \hat{p}_2] = [G_3, \hat{p}_3]$. Combining results from above, we find

$$[\hat{p}_i, G_j] = i\hbar\delta_{ij}m\mathbb{1} \quad (5)$$

where m is a real constant, with a value that cannot be eliminated. One argument proceeds by contradiction. For a spinless particle, suppose $m = 0$, so $[\mathbf{p}, \mathbf{G}] = 0$. At $t = 0$, $[\mathbf{x}, \mathbf{G}] = 0$ (no change in position); thus, components of $\mathbf{G}$ must each be proportional to the identity. This contradicts $[J_i, G_j] = i\hbar\epsilon_{ijk}G_k$ unless $G_i = 0$, so we do not have a consistent set of operators that represents all group operations if $m = 0$. In my attempt to simplify this discussion, I've left out any clear opportunity to show that one *cannot* find a consistent set of commutation relations with $m = 0$, but the author is aware that such arguments exist...

Above, we claimed that $[\hat{x}_i, G_i] = 0$ at $t = 0$, since the boost does not evoke any change in position for this particular instantaneous transformation. It follows that G_i must be a function of $\hat{x}_i$, and furthermore,

$$i\hbar m = [\hat{p}_i, G_i] = -i\hbar \frac{dG_i}{d\hat{x}_i}$$

Thus, $G_i = -m\hat{x}_i$ is one valid solution for the instantaneous boost generator at $t = 0$.

We could go on to find the commutation relations involving H , but those commutators will only apply when the system exhibits full symmetry under Galilean transformations, including spatial translations. We don't expect a particle interacting with an external field to exhibit symmetry under spatial translation.

A spinless particle interacting with external fields

Interactions will modify the time evolution of the quantum state in a manner described by the Schrödinger equation for the appropriate choice of H :

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \quad (6)$$

From the Schrödinger equation, we can readily obtain the Heisenberg equation of motion

$$\frac{d\langle \mathbf{x} \rangle}{dt} = \frac{i}{\hbar} \langle [H, \mathbf{x}] \rangle \quad (7)$$

In light of this result, we define the *velocity operator* as

$$\mathbf{v} = \frac{i}{\hbar} [H, \mathbf{x}] \quad (8)$$

The velocity operator must transform under a unitary boost operation in a way that represents the transformation $\mathbf{v} \rightarrow \mathbf{v} + \mathbf{u}$. Hence,

$$e^{\frac{i}{\hbar} \mathbf{u} \cdot \mathbf{G}} \mathbf{v} e^{-\frac{i}{\hbar} \mathbf{u} \cdot \mathbf{G}} = \mathbf{v} + \mathbf{u}, \quad (9)$$

which is analogous to the “adjoint formula” for rotations in lecture #4. Differentiating both sides with respect to u_i and setting $u_i = 0$ gives

$$[\hat{v}_i, G_j] = i\hbar \delta_{ij} \quad (10)$$

It is true that $\mathbf{v} = \mathbf{p}/m$ solves the commutation relations given by Eqs. (5) and (10), but the solution is not unique. Nonetheless, with this choice for the velocity operator, we can set about finding the Hamiltonian that satisfies Eq. (8). One solution is simply

$$H_0 = \frac{\mathbf{p}^2}{2m}$$

and another solution differs only by the addition of a constant factor multiplying the identity. From the definition of the velocity operator and the solution $\mathbf{v} = \mathbf{p}/m$, we find

$$[H - H_0, \mathbf{x}] = [H, \mathbf{x}] - [H_0, \mathbf{x}] = -i\hbar \mathbf{v} + i\hbar \frac{\mathbf{p}}{m} = 0$$

which means $H - H_0$ commutes with $\mathbf{x}$, so it must be a function of $\mathbf{x}$; let's call it $V(\mathbf{x})$. The

Hamiltonian respecting Eq. (8) and the particular solution $\mathbf{v} = \mathbf{p}/m$ is

$$H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x})$$

As mentioned earlier, $\mathbf{v} = \mathbf{p}/m$ is not a unique solution to Eqs. (5) and (10). Consider the commutator of $\mathbf{v} - \mathbf{p}/m$ with $\mathbf{G}$:

$$[\mathbf{v} - \mathbf{p}/m, \mathbf{G}] = [\mathbf{v}, \mathbf{G}] - [\mathbf{p}/m, \mathbf{G}] = i\hbar - i\hbar = 0$$

Since $\mathbf{G} = -m\mathbf{x}$ for the instantaneous transformation at $t = 0$, the linear combination $\mathbf{v} - \mathbf{p}/m$ commutes with $\mathbf{x}$ and must therefore be a function of $\mathbf{x}$; let's call it $-\mathbf{A}(\mathbf{x})/m$. Thus, the most general form of the velocity operator for a spinless particle is

$$\mathbf{v} = \frac{\mathbf{p} - \mathbf{A}(\mathbf{x})}{m} \quad (11)$$

and if we solve the commutation relation that defines $\mathbf{v}$, we find one possible solution

$$H_1 = \frac{[\mathbf{p} - \mathbf{A}(\mathbf{x})]^2}{2m}$$

By a similar line of reasoning as before, the most general Hamiltonian is

$$H = \frac{[\mathbf{p} - \mathbf{A}(\mathbf{x})]^2}{2m} + V(\mathbf{x}) \quad (12)$$

With this result, we have deduced that the only forms of interaction consistent with invariance under the Galilei group of transformations are a *scalar potential* $V(\mathbf{x})$ and a *vector potential* $\mathbf{A}(\mathbf{x})$. Both of these may be time-dependent. As operators they may be functions of $\mathbf{x}$ but must be independent of $\mathbf{p}$.